

Ayush Kumar

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EDUCATION

National Institute of Technology, Durgapur

Bachelor of Technology in Computer Science

Durgapur, West Bengal, India

Nov. 2022 – May 2026

Eklavya Educational Complex, Palanga

Class XII

Punpun, Patna, Bihar, India

April 2020 – March 2022

Eklavya Educational Complex, Palanga

Class X

Punpun, Patna, Bihar, India

April 2018 – March 2020

TECHNICAL SKILLS

Programming Languages: Python, C/C++, SQL

Frameworks & Libraries: LangChain, FastAPI, Streamlit, Pandas, NumPy, Matplotlib, Scikit-learn, TensorFlow

Databases & Vector Stores: PostgreSQL, SQLite, FAISS

Core Concepts: Data Structures & Algorithms, Machine Learning, Deep Learning, Retrieval-Augmented Generation (RAG), REST APIs

Developer Tools: Git, GitHub, Docker, VS Code, Visual Studio, IntelliJ IDEA

EXPERIENCE

Machine Learning Intern

CODTECH IT SOLUTIONS

July 2025 – Present

India

- Role: Software Development Intern Duration: July 2025 – Present Mode: Remote (Work-from-home).
- Description: Received an official internship offer from CODTECH IT SOLUTIONS to contribute to real-world software development projects. The internship emphasizes hands-on experience in full-stack development, algorithmic problem-solving, and deployment workflows. Responsibilities include working on industrial-grade applications, collaborating with the core tech team, and delivering tasks within defined deadlines. The program also offers placement support, resume-building guidance, and interview preparation resources.
- LinkedIn Link: <https://url-shortener.me/7ZLQ>

PROJECTS

Document RAG (Retrieval-Augmented Generation) | *Python, Streamlit, LangChain, FAISS, Hugging Face, Groq*

- Developed an AI-powered document question-answering application that allows users to upload PDF, DOCX, TXT, and CSV files and ask natural language questions about their content.
- Implemented a RAG pipeline using LangChain with Recursive Character Text Splitting, Hugging Face BGE embeddings, and FAISS vector database for efficient semantic retrieval.
- Integrated the Groq Llama-3.3-70B model to generate context-aware answers based only on the retrieved document content, reducing hallucinations.
- Designed an interactive Streamlit interface with Maximum Marginal Relevance (MMR) retrieval and support for multiple document formats to enhance the user experience.
- GitHub Repository: https://github.com/AyushKumar1080/Doc_RAG

Book Recommender System | *Python, Flask, HTML*

- Developed a content-based recommendation engine that suggests books similar to a user's selection using cosine similarity on TF-IDF vectorized book descriptions.
- Built with Flask and styled using Bootstrap, the system offers a dynamic, responsive UI that updates recommendations in real time.
- Leveraged Jinja2 templating for efficient content rendering and Pandas for backend data manipulation.
- GitHub Repository: <https://github.com/AyushKumar1080/Movies-App-New.git>